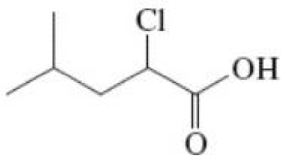
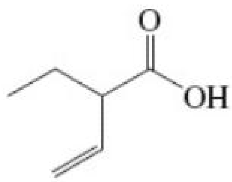
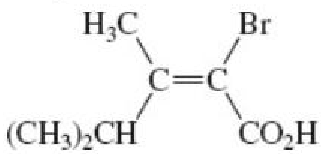
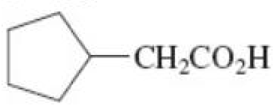
Problems

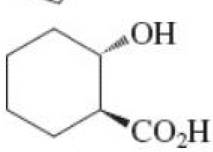
27. Name (IUPAC or *Chemical Abstracts* system) or draw the structure of each of the following compounds.

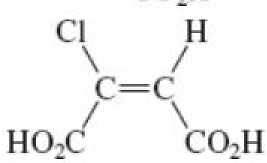
a. 

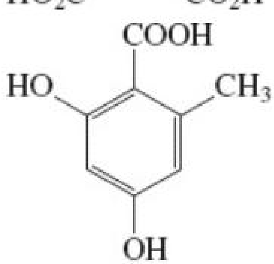
b. 

c. 

d. 

e. 

f. 

g. 

***o*-Orsellinic acid**

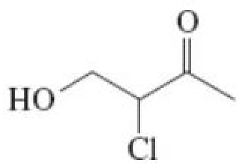
h. 

Phthalic acid

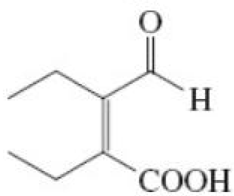
(i) 4-Aminobutanoic acid (also known as GABA, a critical participant in brain biochemistry); **(j)** *meso*-2,3-dimethylbutanedioic acid; **(k)** 2-oxopropanoic acid (pyruvic acid); **(l)** *trans*-2-formylcyclohexanecarboxylic acid; **(m)** (*Z*)-3-phenyl-2-butenoic acid; **(n)** 1,8-naphthalenedicarboxylic acid.

28. Name each of the following compounds according to IUPAC or *Chemical Abstracts.* Pay attention to the choice of the main chain and the order of precedence of the functional groups.

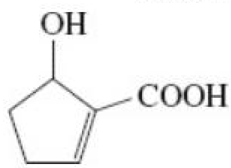
a.



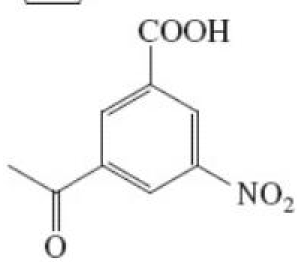
b.



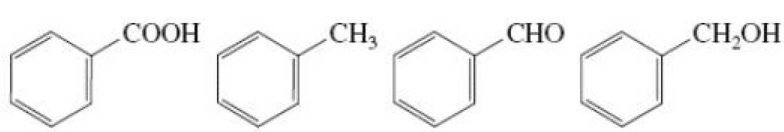
c.



d.



29. Rank the group of molecules shown here in decreasing order of boiling point and solubility in water. Explain your answers.



30. Rank each of the following groups of organic compounds in order of decreasing acidity.

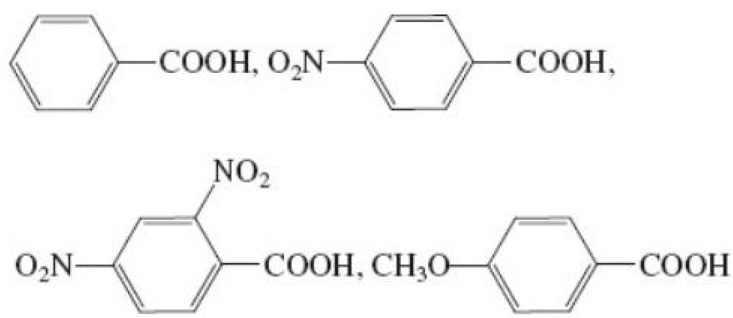
a. $CH_3CH_2CO_2H$, $CH_3C(=O)CH_2OH$, $CH_3CH_2CH_2OH$

b. BrCH2CO2H,ClCH2CO2H,FCH2CO2H
$BrCH_2CO_2H, ClCH_2CO_2H, FCH_2CO_2H$

c. $CH_3CHClCH_2CO_2H$, $ClCH_2CH_2CH_2CO_2H$, $CH_3CH_2CHClCO_2H$

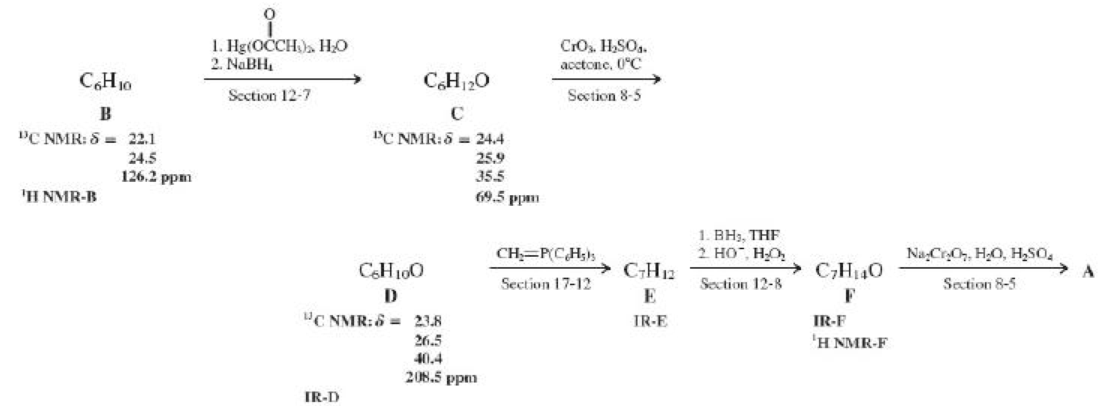
d. CF3CO2H,CBr3CO2H,(CH3)3CCO2H
$CF_3CO_2H, CBr_3CO_2H, (CH_3)_3CCO_2H$

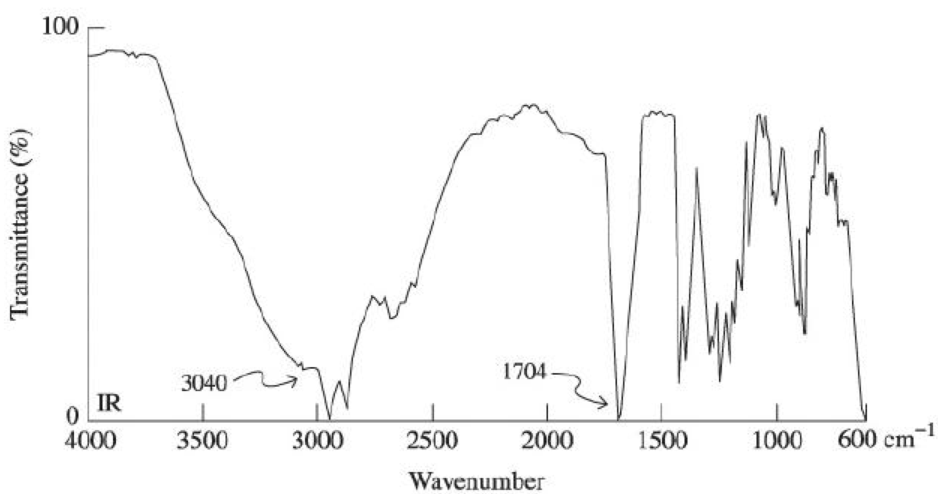
e.



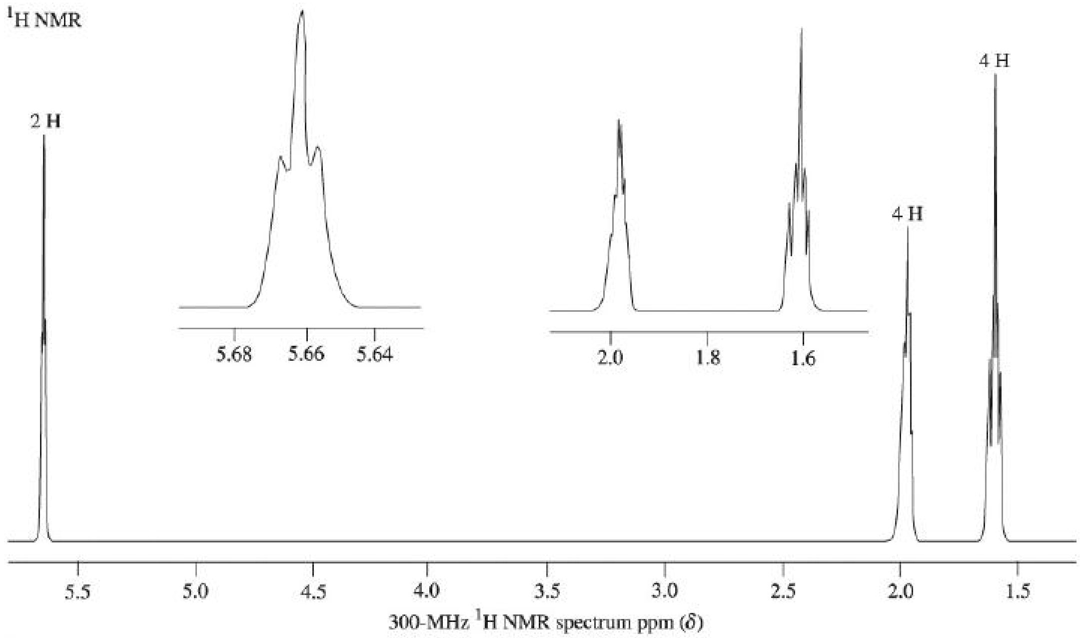
31. Propose a structure for a compound that displays the spectroscopic data that follow. The molecular ion in the mass spectrum appears at m/z=116 $m/z = 116$. IR: ṽ=1710 (s), $\mathrm{IR}: \tilde{v} = 1710\ (s)$, 3000 (s, broad) cm−1; 3000 (s, broad) cm^{-1}; 1H NMR: δ=0.94 (t, J=7.0 Hz, 6 H), $^1\mathrm{H\ NMR}: \delta = 0.94$ (t, $J = 7.0$ Hz, 6 H), 1.59 (m, 4 H), 1.59 (m, 4 H), 2.36 (quin, J=7.0 Hz, 1 H), 2.36 (quin, $J = 7.0$ Hz, 1 H), 12.04 (broad s, 1 H) ppm; 12.04 (broad s, 1 H) ppm; 13C NMR: δ=11.7, $^{13}\mathrm{C\ NMR}: \delta = 11.7$, 24.7, 24.7, 48.7, 48.7, 183.0 ppm. 183.0 ppm.

32. **(a)** An unknown A has the formula C7H12O2 $C_7H_{12}O_2$ and infrared spectrum A (below). To which class does this compound belong? **(b)** Use the other spectra (NMR-B, p. 961, and F, p. 962; IR-D, E, and F, pp. 961-962) and spectroscopic and chemical information in the reaction sequence to determine the structures of A and the other unknown substances B through F. References are made to relevant sections of earlier chapters, but do not look them up before you have tried to solve the problem without the extra help. **(c)** Another unknown compound, G, has the formula C8H14O4$C_8H_{14}O_4$ and the NMR and IR spectra labeled G (p. 962-963). Propose a structure for this molecule. **(d)** Compound G may be readily synthesized from B. Propose a sequence that accomplishes this efficiently. **(e)** Propose a completely different sequence from that shown in part (b) for the conversion of C into A. **(f)** Finally, construct a synthetic scheme that is the reverse of that shown in part (b); namely, the conversion of A into B.



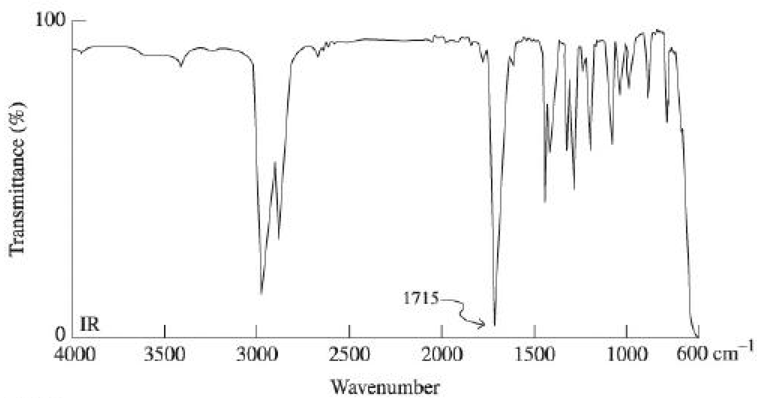


IR-A



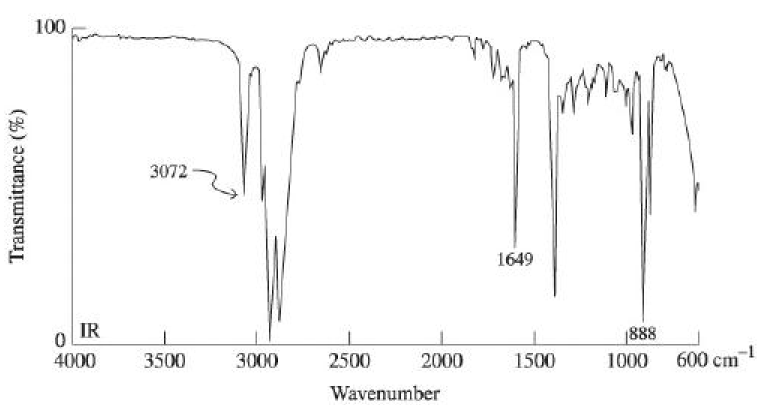
1H NMR
2 H
4 H
4 H
5.68
5.66
5.64
2.0
1.8
1.6
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
300-MHz 1H NMR spectrum ppm (δ)

B



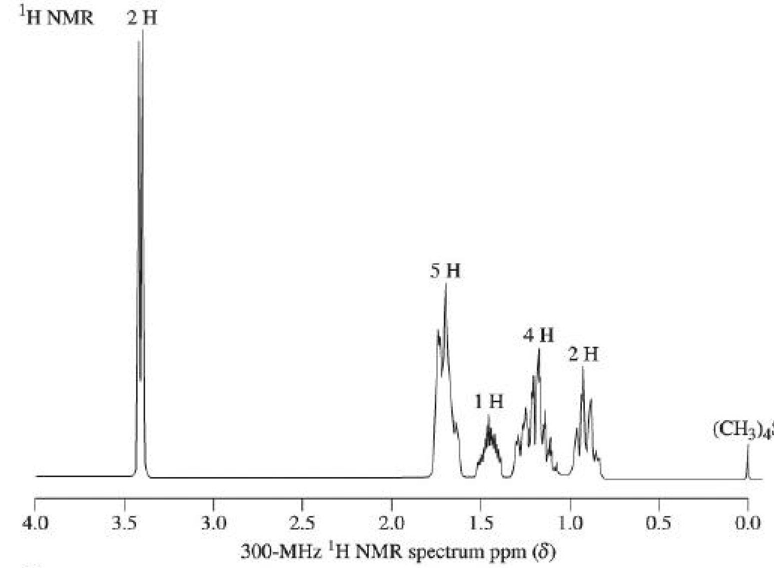
100
0
Transmittance (%)
IR
1715
4000
3500
3000
2500
2000
1500
1000
600 cm−1
Wavenumber

IR-D



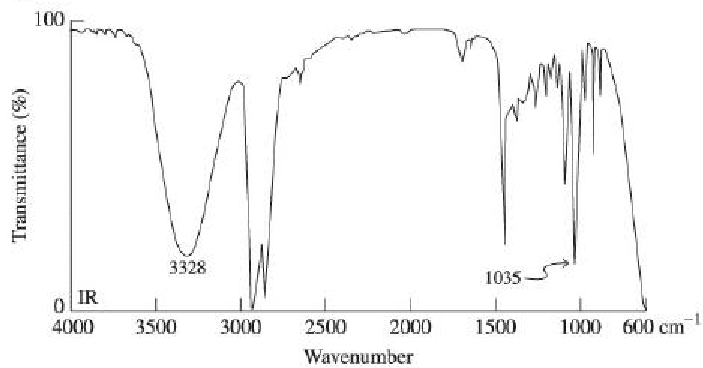
100
0
Transmittance (%)
IR
3072
1649
888
4000
3500
3000
2500
2000
1500
1000
600 cm−1
Wavenumber

IR-E



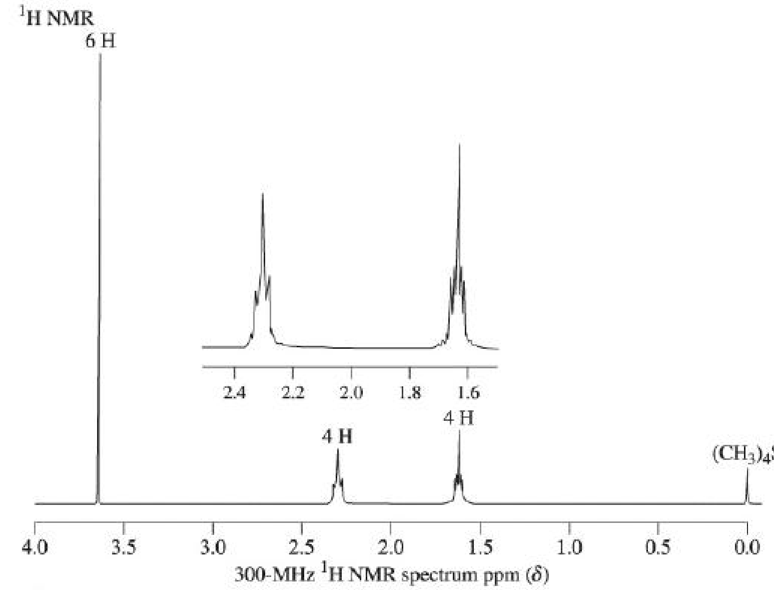
^{1}H NMR
2 H
5 H
4 H
2 H
1 H
$(CH_3)_4S$
4.0 3.5 3.0 2.5 2.0 1.5 1.0 0.5 0.0
300-MHz ^{1}H NMR spectrum ppm (δ)

F



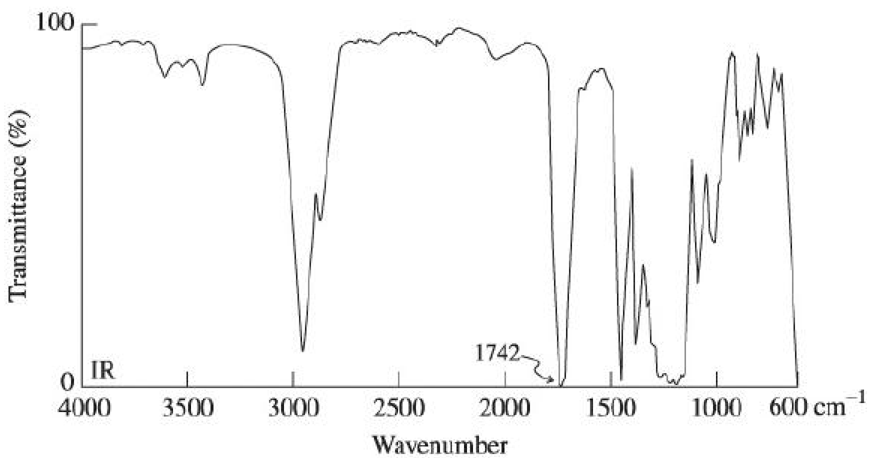
100
Transmittance (%)
3328
1035
0
IR
4000 3500 3000 2500 2000 1500 1000 600 cm^{-1}
Wavenumber

IR-F



^{1}H NMR
6 H
2.4 2.2 2.0 1.8 1.6
4 H
4 H
$(CH_3)_4Si$
4.0 3.5 3.0 2.5 2.0 1.5 1.0 0.5 0.0
300-MHz ^{1}H NMR spectrum ppm (δ)

G

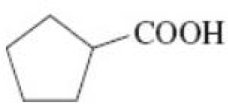


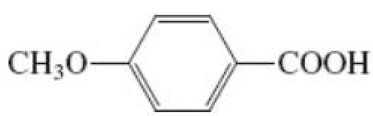
IR-G

33. Give the products of each of the following reactions.

a. (CH3)2CHCH2CO2H+SOCl2→ $(CH_3)_2CHCH_2CO_2H + SOCl_2 \rightarrow$

b. (CH3)2CHCH2CO2H+CH3COBr→ $(CH_3)_2CHCH_2CO_2H + CH_3COBr \rightarrow$

c.  $+ \ CH_3CH_2OH \xrightarrow{H^+}$

d.  $+ \ NH_3 \longrightarrow$

e. Product of (d), heated strongly

f. Phthalic acid [Problem 27, part (h)], heated strongly

34. When 1,4- and 1,5-dicarboxylic acids, such as butanedioic (succinic) acid (Section 19-8), are treated with SOCl2$SOCl_2$ or PBr3PBr_3 in attempted preparations of the diacyl halides, the corresponding cyclic anhydrides are obtained. Explain mechanistically.

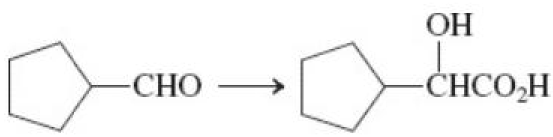
35. Reaction review. Without consulting the Reaction Road Map on p. 958, suggest reagents to convert each of the following starting materials into hexanoic acid: **(a)** hexanal; **(b)** methyl hexanoate; **(c)** 1-bromopentane; **(d)** 1-hexanol; **(e)** hexanenitrile.

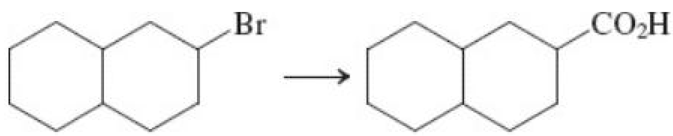
36. Reaction review II. Without consulting the Reaction Road Map on p.

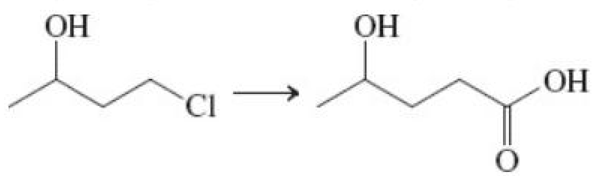
959, suggest reagents to convert hexanoic acid into each of the following compounds: **(a)** 1-hexanol; **(b)** hexanoic anhydride; **(c)** hexanoyl chloride; **(d)** 2-bromohexanoic acid; **(e)** ethyl hexanoate; **(f)** hexanamide.

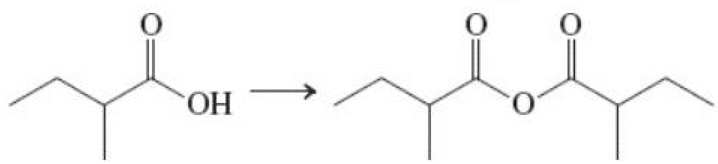
37. Fill in suitable reagents to carry out the following transformations.

a. (CH3)2CHCH2CHO→(CH3)2CHCH2CO2H
$(CH_3)_2CHCH_2CHO \rightarrow (CH_3)_2CHCH_2CO_2H$

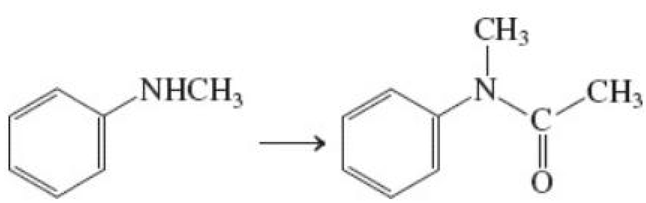
b.


c.


d.


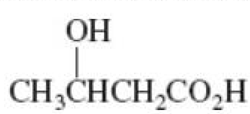
e.


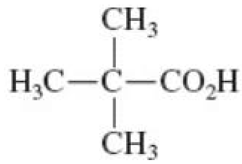
f. $(CH_3)_3CCO_2H \longrightarrow (CH_3)_3CCO_2CH(CH_3)_2$

g.


38. Propose syntheses of each of the following carboxylic acids that employ at least one reaction that forms a carbon-carbon bond.

a. CH3CH2CH2CH2CH2CH2CO2H$CH_3CH_2CH_2CH_2CH_2CH_2CO_2H$

b.


c. 

39. **(a)** Write a mechanism for the esterification of propanoic acid with 18O $^{18}\mathrm{O}$-labeled ethanol. Show clearly the fate of the 18O$^{18}\mathrm{O}$ label. **(b)** Acid-catalyzed hydrolysis of an unlabeled ester with 18O$^{18}\mathrm{O}$-labeled water (H218O)$(\mathrm{H_2}{}^{18}\mathrm{O})$ leads to incorporation of some 18O$^{18}\mathrm{O}$ into *both* oxygens of the carboxylic acid product. Explain by a mechanism. (**Hint:** You must use the fact that all steps in the mechanism are reversible.)

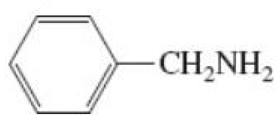
40. Give the products of reaction of propanoic acid with each of the following reagents.

a. SOCl2$\mathrm{SOCl_2}$

b. PBr3$\mathrm{PBr_3}$

c. CH3CH2COBr+pyridine$\mathrm{CH_3CH_2COBr} + \text{pyridine}$

d. (CH3)2CHOH+HCl$(\mathrm{CH_3})_2\mathrm{CHOH} + \mathrm{HCl}$

e. 

f. Product of (e), heated strongly

g. LiAlH4,$\mathrm{LiAlH_4}$, then H+,$\mathrm{H^+}$, H2O$\mathrm{H_2O}$

h. Br2,P$\mathrm{Br_2}$, P

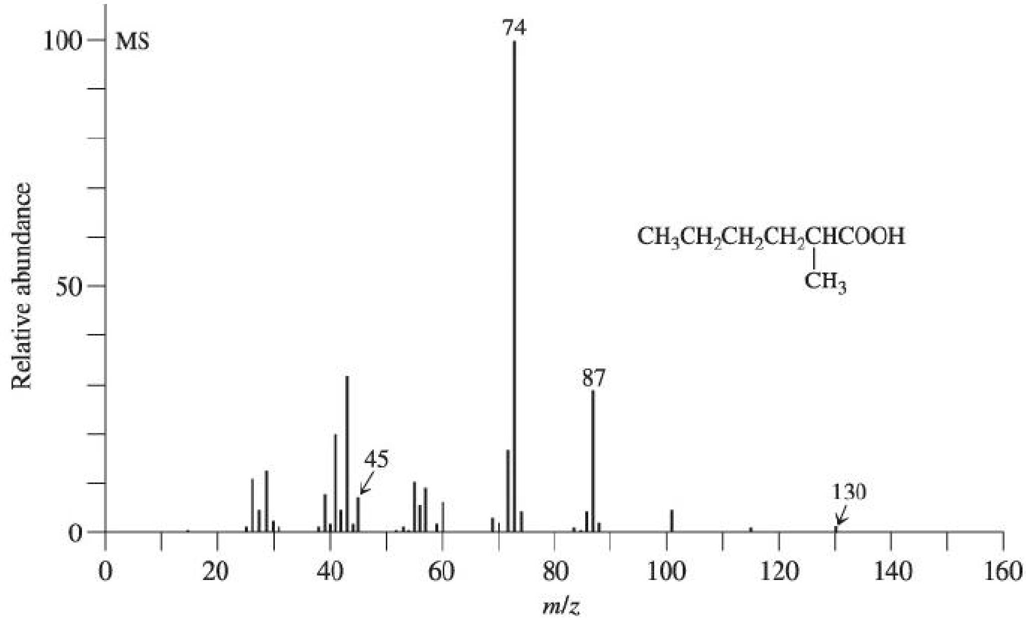
41. Give the product of reaction of cyclopentanecarboxylic acid with each of the reagents in Problem 40.

42. **CHALLENGE** When methyl ketones are treated with a halogen in the presence of base, the three hydrogen atoms on the methyl carbon are replaced to give a CX3$\mathrm{CX_3}$-substituted ketone (Section 18-3). This product is not stable indefinitely under the basic conditions and proceeds to react further with hydroxide, ultimately furnishing the carboxylic acid (as its conjugate base) and a molecule of HCX3,$\mathrm{HCX_3}$, which has the common name haloform (i.e., chloroform, bromoform, and iodoform, for X=Cl,$X = Cl$, Br,Br, and I,I, respectively). For example:

$$R\overset{\overset{:O:}{\|}}{C}CBr_3 \;+\; {}^{-}:\ddot{\underset{\cdot\cdot}{O}}H \;\longrightarrow\; R\overset{\overset{:O:}{\|}}{C}\ddot{\underset{\cdot\cdot}{O}}:^{-} \;+\; HCBr_3$$

Propose a series of mechanistic steps to convert the tribromoketone into the carboxylate. What is the leaving group? Why do you think this species is capable of acting as a leaving group in this process?

43. Interpret the labeled peaks in the mass spectrum of 2-methylhexanoic acid (MS-H).



MS-H

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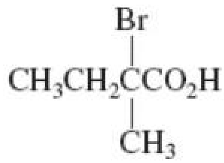
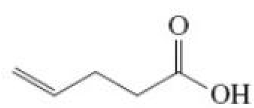
44. Suggest a preparation of hexanoic acid from pentanoic acid.

45. Give reagents and reaction conditions that would allow efficient conversion of 2-methylbutanoic acid into **(a)** the corresponding acyl chloride; **(b)** the corresponding methyl ester; **(c)** the corresponding ester with 2-butanol; **(d)** the anhydride; **(e)** the NN-methylamide;

f. $CH_3CH_2\underset{}{\overset{\overset{CH_3}{|}}{C}}HCH_2OH$

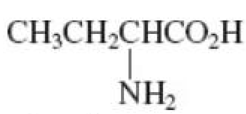
g. 

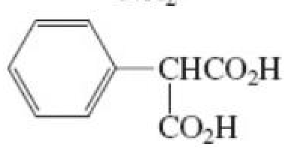
46. Treatment of 4-pentenoic acid (below) with Br2Br_2 in the presence of dilute aqueous base yields a nonacidic material with the formula C5H7BrO2$C_5H_7BrO_2$. **(a)** Suggest a structure for this compound and a mechanism for its formation. **(b)** Can you find a second, isomeric product whose formation is also mechanistically reasonable? **(c)** Discuss the issues that contribute to determining which of the two is the major product in this reaction. (**Hint:** Review Section 12-6.)

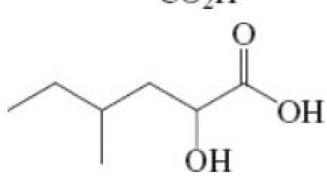


4-Pentenoic acid

47. Show how the Hell-Volhard-Zelinsky reaction might be used in the synthesis of each of the following compounds, beginning in each case with a simple monocarboxylic acid. Write detailed mechanisms for all the reactions in *one* of your syntheses.

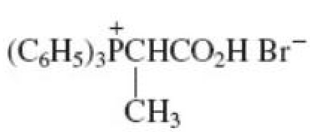
a. 

b. 

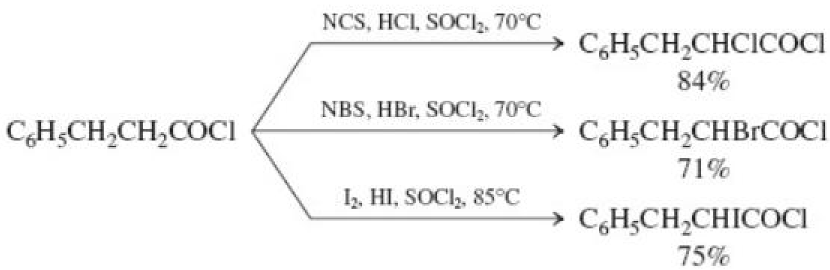
c. 

d. $HO_2CCH_2SSCH_2CO_2H$

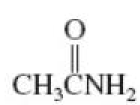
e. $(CH_3CH_2)_2NCH_2CO_2H$

f. 

48. The Hell-Volhard-Zelinsky reaction produces only bromocarboxylic acids. However, modified procedures have been developed to convert acyl chlorides into αα-chloro- and αα-bromoacyl chlorides by reaction with NN-chloro- and NN-bromobutanimide (NN-chloro- and NN-bromosuccinimide, NCS and NBS: Section 14-2), respectively. Reaction with I2I_2 gives αα-iodoacyl chlorides. (SOCl2$SOCl_2$ is used as the solvent to maintain the acyl chloride functional group.) Suggest a mechanism for any one of these processes.

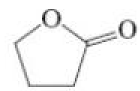


49. **(a)** (4-Nitrophenyl)ethanenitrile (Section 19-6, p. 925) is useful as an acid-base indicator in the high pH range. It is yellow below its pKapK_a of 13.4 and red above. Using the principles that govern acidity, identify the most acidic hydrogen(s) in the molecule, draw the structure of its conjugate base, and provide an explanation for its acidity. **(b)** Draw resonance forms to explain the stability of protonated propenoic acid (Section 19-3, p. 919), a common fragment observed in the mass spectra of longer chain carboxylic acids. **(c)** How would you expect the acidity of acetamide to compare with that of acetic acid? With that of acetone? Which protons in acetamide are the most acidic? Where would you expect acetamide to be protonated by very strong acid?



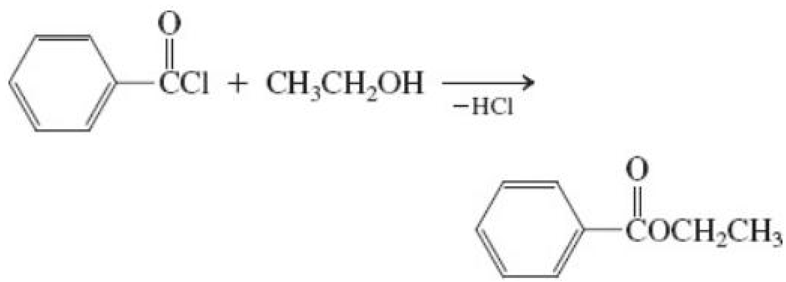
Acetamide

50. Attempted CrO3CrO_3 oxidation of 1,4-butanediol to butanedioic acid results in significant yields of "γ-"γ-butyrolactone." Explain mechanistically.



γ-Butyrolactone

51. Following the general mechanistic scheme given in Section 19-7, write detailed mechanisms for each of the following substitution reactions. (*Note:* These transformations are part of Chapter 20, but try to solve the problem without looking ahead.)

a. 

b. $CH_3\overset{\overset{\large O}{\|}}{C}NH_2 + H_2O \xrightarrow{H^+} CH_3\overset{\overset{\large O}{\|}}{C}OH + \overset{+}{N}H_4$

52. Suggest structures for the products of each reaction in the following synthetic sequence.